

Research Report: Analysis of Housing Materials for BEV Auxiliary Motors and Gearboxes

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Objective: This report provides a comprehensive analysis of the distribution and selection of aluminum versus steel for the housings of Battery Electric Vehicle (BEV) auxiliary motors and gearboxes. It examines the material choices based on motor type and vehicle segment, offering a detailed evaluation of the trade-offs between these two primary materials.

Executive Summary

The selection of housing materials for auxiliary motors and gearboxes in Battery Electric Vehicles is a critical engineering decision that directly influences vehicle efficiency, performance, and overall cost. Our analysis reveals a clear and dominant trend toward the use of **aluminum alloys** over steel for these applications [2, 5]. This preference is driven by aluminum's superior combination of low weight, excellent thermal conductivity, and advanced manufacturing flexibility through die-casting [5, 9]. These properties are paramount in the context of BEVs, where maximizing driving range through weight reduction and ensuring optimal performance through effective thermal management are primary objectives [5].

While steel remains a viable and cost-effective material for general-purpose industrial motors and retains a critical, non-negotiable role in the magnetic circuits of certain motor designs [1, 4], its application for the external housings of BEV components is limited. The significant weight penalty and inferior heat dissipation capabilities of steel are considerable disadvantages in the highly optimized environment of a modern electric vehicle [5, 9]. For BEV gearboxes, a similar trend is observed, with a marked shift away from traditional cast iron and steel towards lightweight aluminum to reduce vehicle mass [11]. Although specific material choices are not explicitly detailed by vehicle segment in available data, the overarching industry pursuit of efficiency suggests that this preference for aluminum is pervasive across all segments, from compact city cars to large premium vehicles. This report details the technical and economic factors underpinning these material selection trends, providing a granular analysis of the trade-offs involved.

Analysis of BEV Auxiliary Motor Housings

The housing of an electric motor serves multiple essential functions: it provides structural support and protection for the internal components, aids in thermal management by dissipating waste heat, and in some designs, forms part of the motor's magnetic circuit [2, 4]. In the context of BEVs, the material chosen for this housing has profound implications for the vehicle's performance and efficiency. The industry has overwhelmingly converged on aluminum as the material of choice for the housings of auxiliary motors [2, 5].

The Predominance of Aluminum in BEV Motor Housings

Aluminum alloys have become the standard for modern BEV motor housings due to a confluence of properties that align perfectly with the demands of vehicle electrification [2, 5, 9]. The primary drivers

for its adoption are its low density, high thermal conductivity, and the design freedom afforded by high-pressure die casting [5, 9].

A key objective in BEV design is the reduction of overall vehicle mass to extend driving range. Aluminum, with a density approximately one-third that of steel, offers a substantial weight-saving advantage [5, 8]. Every kilogram of mass removed from the vehicle contributes to improved energy efficiency, making the lightweight nature of aluminum a critical asset [5]. This reduction in the motor's weight also has secondary benefits, such as reducing inertia in dynamic applications, which can lead to faster response times and more precise control [8].

Equally important is thermal management. Electric motors generate significant heat, particularly in the stator windings, during operation. This heat must be efficiently removed to prevent the degradation of winding insulation, maintain motor efficiency, and ensure long-term reliability [2, 5]. Aluminum is an excellent thermal conductor, with a conductivity of approximately 150-200 W/mK, which is three to four times higher than that of carbon steel (around 45 W/mK) or cast iron (around 50 W/mK) [2]. This allows an aluminum housing to function as a highly effective heat sink, drawing thermal energy away from the motor's core and transferring it to the ambient air or a liquid cooling system [2, 5].

Furthermore, the manufacturing process most commonly used for aluminum housings—high-pressure die casting—enables the creation of highly complex and integrated designs [5, 9]. Engineers can design housings with intricate cooling fins, sophisticated internal liquid cooling channels, and mounting bosses directly integrated into a single component [3, 5]. This “net-shape” manufacturing capability minimizes the need for subsequent machining operations, reducing production time and cost [5]. The ability to create thin-walled yet strong structures, with wall thicknesses down to 2-3 mm, further contributes to lightweighting without compromising structural integrity [9]. Common alloys used for this purpose, such as those from the Al-Si-Cu family (e.g., A380, ADC12), provide an optimal balance of castability, mechanical strength, and thermal performance [3].

The Specialized Role of Steel in Motor Housings

Despite the dominance of aluminum, steel continues to play a crucial, albeit more specialized, role in electric motors. Its use as an external housing material in BEVs is limited due to its significant weight and poor thermal conductivity [5, 9]. However, its magnetic properties are indispensable for the motor's core function [1, 4].

In many electric motor designs, particularly traditional DC motors, the housing itself serves as the **magnetic yoke** [1, 4]. The yoke is a fundamental part of the magnetic circuit, providing a path for the magnetic flux between the rotor and stator [1]. This function requires a ferromagnetic material, for which steel and iron are the only practical choices [1]. Using a non-ferromagnetic material like aluminum for the yoke would severely impede the magnetic flux, drastically reducing the motor's torque and efficiency [1]. Therefore, in motor architectures where the housing is integral to the magnetic circuit, steel or iron is not just a choice but a necessity [1].

This distinction is critical. While the external, protective enclosure of a motor can be made of aluminum to save weight and dissipate heat, the internal magnetic path must be preserved. In modern BEV motor design, this is often achieved through a hybrid approach. For instance, a motor might feature a thin steel or iron shell pressed into a larger, finned aluminum housing [1]. This design leverages the magnetic properties of steel for the yoke while capitalizing on the lightweight and thermal benefits of the external aluminum structure. Similarly, even on motors with a main body made of steel, components that are not part of the magnetic circuit, such as the endplates or end bells, are frequently made from die-cast aluminum to reduce weight [1, 4].

Steel remains the “practical workhorse” for general-purpose industrial motors, such as standardized NEMA and IEC frame motors used in applications like pumps and conveyors [2]. In these contexts, extreme lightweighting is not a priority, and the durability, strength, and lower raw material cost of fabricated steel make it an economical and reliable choice [2].

Material Selection by Motor Type

The choice between aluminum and steel is heavily influenced by the specific type of auxiliary motor and its operating principles.

For the majority of auxiliary motors found in modern BEVs, which are typically Permanent Magnet Synchronous Motors (PMSM) or AC Induction Motors, the external housing’s primary roles are structural and thermal. In these designs, the magnetic circuit is contained within the stator and rotor laminations, and the external housing does not serve as the magnetic yoke [1, 4]. Consequently, the material selection is driven by the need for weight reduction and heat dissipation, making aluminum the unequivocal best choice [5, 9]. Its ability to be cast into complex shapes with integrated cooling features allows for the creation of compact, power-dense motors that are essential for vehicle packaging and efficiency [8].

The situation is different for certain types of **DC motors**. As previously discussed, the housing of a brushed or brushless DC motor often functions as the magnetic yoke [1]. In these cases, a steel or iron housing is required to ensure proper magnetic flux and torque production [1]. Replacing this with an aluminum housing would fundamentally compromise the motor’s performance. Therefore, for auxiliary applications that utilize this type of DC motor, steel remains the necessary material for the main body, although aluminum may be used for non-magnetic components like endplates [1, 4].

For smaller auxiliary motors like **stepper motors**, which are used for functions such as controlling valves or adjusting mirrors, the material trends are similar. While specific distribution percentages are not available, performance-driven applications favor aluminum for its superior heat dissipation, which is crucial for maintaining precision and preventing overheating [7]. For more standard, less demanding applications where cost is the primary driver, steel remains a common and practical option [2]. It is important to note that regardless of the housing material, the internal stator and rotor laminations of a stepper motor are made from silicon steel to optimize magnetic properties and reduce core losses [7].

Analysis of BEV Gearbox Housings

The gearbox in a BEV, though simpler than its internal combustion engine counterpart, requires a housing that is strong, lightweight, and acoustically dampened. The material selection for these housings follows a trajectory similar to that of motor housings, with a clear industry-wide movement toward lightweight materials.

Historically, automotive gearbox housings, especially in heavy-duty vehicles, were made from robust materials like **grey cast iron** (e.g., HT200, FG260) or structural steel [11]. These materials offer excellent strength and rigidity. However, they are also extremely heavy. For example, analysis of a heavy vehicle truck transmission showed a grey cast iron housing weighing nearly 55 kg and a structural steel equivalent weighing almost 60 kg [11].

In the context of BEVs, such weight is untenable. The push for efficiency has driven a transition to **aluminum alloys** for gearbox housings [11]. The same study that analyzed heavy vehicle gearboxes found that an equivalent housing made from an aluminum alloy weighed just over 21 kg—a weight reduction of over 60% compared to its iron and steel counterparts [11]. This dramatic mass reduction

directly translates to improved vehicle range and is a primary reason for aluminum’s adoption in BEV gearbox applications [11].

While the available research does not provide a detailed breakdown of gearbox housing materials by specific European vehicle segments (A, B, C, D, E, F), the universal goal of maximizing efficiency strongly suggests that lightweight aluminum housings are the standard across all passenger BEV segments. The benefits of weight reduction are segment-agnostic, and the manufacturing advantages of aluminum die-casting apply equally to small city cars and large SUVs.

Beyond weight, BEV gearbox housings present unique challenges related to Noise, Vibration, and Harshness (NVH) and thermal management. The high-frequency sounds produced by electric motors and gear whine can be more perceptible in the absence of a loud internal combustion engine. Lightweight designs can sometimes exacerbate NVH issues. Consequently, research is focused on optimizing the design of aluminum housings with features like ribbed panels to improve stiffness and acoustic performance. Emerging solutions also include the exploration of composite materials, which offer potential for both further weight reduction and enhanced vibration damping.

Thermal management is also a consideration. Some advanced BEV architectures are exploring integrated thermal systems, or “mono fluid circuits,” where a single fluid is used for both gearbox lubrication and the cooling of power electronics and the electric motor [13]. In such a system, the thermal properties of the gearbox housing material become part of the overall thermal management strategy, further underscoring the importance of material selection [13].

Summary Tables of Material Comparison

To consolidate the key differences between aluminum and steel for these applications, the following tables summarize their properties and suitability.

Table 1: Comparative Analysis of Material Properties

Property	Aluminum Alloy	Carbon Steel
Thermal Conductivity (W/mK)	Excellent (approx. 150-200) [2]	Poor (approx. 45) [2]
Density (g/cm³)	Low (approx. 2.7) [8]	High (approx. 7.85)
Relative Weight	Lightweight (approx. 1/3 of steel) [5, 8]	Heavy
Magnetic Properties	Non-ferromagnetic [1]	Ferromagnetic [1]
Corrosion Resistance	Excellent (forms natural protective oxide layer) [4, 9]	Poor (requires protective coatings to prevent rust) [4, 9]
Primary Manufacturing Process	High-Pressure Die Casting [2, 9]	Fabrication (rolling, stamping, welding) [2, 9]
Design Complexity	High (allows for integrated, complex geometries) [5, 9]	Low to Moderate (limited to simpler forms)

Table 2: Application Suitability for BEV Components

Component	Preferred Material	Key Rationale and Considerations
Motor Housing (General BEV)	Aluminum	Primary Choice. Selected for superior thermal management [2, 8], significant weight reduction for improved vehicle range [5], and design flexibility via die-casting to integrate cooling features [3, 5]. The housing is typically not part of the magnetic circuit [1, 4].
Motor Housing (DC Motor Yoke)	Steel	Required. Selected when the housing must function as the magnetic yoke [1]. Steel’s ferromagnetic properties are essential for motor torque and efficiency [1]. Weight and thermal performance are secondary to this magnetic function.
Gearbox Housing	Aluminum	Primary Choice. Selected for substantial weight reduction to enhance vehicle efficiency and range [11]. Replaces traditional heavy materials like cast iron and steel [11]. Design must also account for NVH performance.

Detailed Trade-Off Analysis: Aluminum vs. Steel

The decision between aluminum and steel for BEV motor and gearbox housings involves a multi-faceted trade-off analysis encompassing weight, thermal performance, cost, and mechanical properties.

Weight and Efficiency: This is the most significant differentiator in the context of BEVs. Aluminum is the undisputed winner, offering a weight reduction of up to 60-70% compared to steel for a component of the same volume [11]. This mass reduction is a direct contributor to extending the vehicle’s driving range, a paramount goal for all electric vehicle manufacturers [5]. The lower weight of steel is a fundamental disadvantage that is difficult to overcome.

Thermal Management and Performance: Aluminum’s thermal conductivity is three to four times greater than that of steel [2]. This allows aluminum housings to dissipate heat far more effectively, enabling motors to operate at higher power densities without overheating [8]. This superior thermal per-

formance can lead to more compact and powerful motor designs, extending the motor's service life by protecting insulation from thermal degradation [8]. A steel housing is a comparatively poor heat sink, which could necessitate larger, heavier, or more complex auxiliary cooling systems, adding cost and negating some of steel's initial material cost advantage [9].

Cost Analysis: The cost comparison is nuanced and extends beyond the raw material price. While steel as a commodity is generally cheaper than aluminum, the total cost of the finished component and its impact on the overall vehicle system must be considered. For complex components, aluminum die-casting can be highly cost-effective. It allows for the consolidation of multiple parts into a single integrated unit, reducing assembly time, labor costs, and potential points of failure [9]. Steel fabrication, involving stamping, welding, and assembly of multiple pieces, can incur higher labor and tooling costs for complex geometries [9].

Most importantly, in a BEV, the cost analysis must be holistic. The weight savings achieved by using aluminum for motor and gearbox housings can enable the use of a smaller, lighter, and therefore less expensive battery pack to achieve the same vehicle range [10]. Research has indicated that while an aluminum-intensive vehicle body may have higher manufacturing costs, the savings in subsystems like the battery and motors can be substantial, potentially making the aluminum-based design more cost-effective at the system level [10].

Strength, Durability, and Corrosion: While cast iron and steel are known for their exceptional strength and rigidity, making them ideal for heavy-duty industrial machinery, modern aluminum alloys, when designed correctly with features like reinforcing ribs, provide more than sufficient strength and impact resistance for automotive applications [8]. Furthermore, aluminum possesses inherent corrosion resistance due to the formation of a protective aluminum oxide layer on its surface [4, 9]. Steel, conversely, is highly susceptible to rust and requires extensive and durable coatings for protection, especially in the harsh underbody environment of a vehicle [4, 9].

Conclusion

The analysis of housing materials for BEV auxiliary motors and gearboxes reveals a decisive industry preference for aluminum alloys over steel. This trend is fundamentally driven by the unique and demanding requirements of electric vehicles, where efficiency and performance are inextricably linked to weight and thermal management.

Aluminum's combination of low density, high thermal conductivity, and the design flexibility offered by die-casting makes it the optimal material for creating lightweight, power-dense, and thermally efficient components. Its use directly contributes to extending vehicle range, a key metric for BEV competitiveness.

Steel's role has become highly specialized. While its lower raw material cost and strength are valuable in certain industrial contexts, its weight and poor thermal properties make it largely unsuitable for the external housings of BEV motors and gearboxes. Its primary and critical application in modern motors is now confined to its function as a ferromagnetic material within the magnetic circuit, a role that aluminum cannot fulfill. In these cases, hybrid designs that combine an internal steel yoke with an external aluminum housing are becoming common, leveraging the best properties of both materials.

For BEV gearboxes, the transition from traditional cast iron and steel to lightweight aluminum is a clear and established trend aimed at reducing overall vehicle mass. Across all vehicle segments, the pursuit of greater efficiency dictates the use of the lightest practical materials. Therefore, aluminum is positioned to remain the dominant material for these critical BEV components for the foreseeable future.

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